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Experiment 6

Application of K-Means and Hierarchical Clustering on Iris Dataset

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1. Dataset Source

Dataset: **Iris Dataset**
Source: UCI Machine Learning Repository
Loaded using sklearn library.

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2. Dataset Description

The Iris dataset contains measurements of iris flowers.

📊 Dataset Size

- 150 samples
- 4 features

📌 Features

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

The dataset originally has 3 classes, but in this experiment, clustering is performed without using class labels.

3.Mathematical Formulation

K-Means

$$\min \sum_{i=1}^{k} \sum_{x \in C_i} ||x - \mu_i||^2$$

Minimizes within-cluster variance.

Hierarchical Clustering

Agglomerative approach:

- Start with each data point as a cluster
- Merge closest clusters
- Continue until one cluster remains

Ward method minimizes variance between clusters.

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4. Algorithm Limitations

K-Means

- Requires pre-defined k
- Sensitive to initial centroid
- Works best for spherical clusters

Hierarchical Clustering

- Computationally expensive for large datasets
- Once merged, clusters cannot be split

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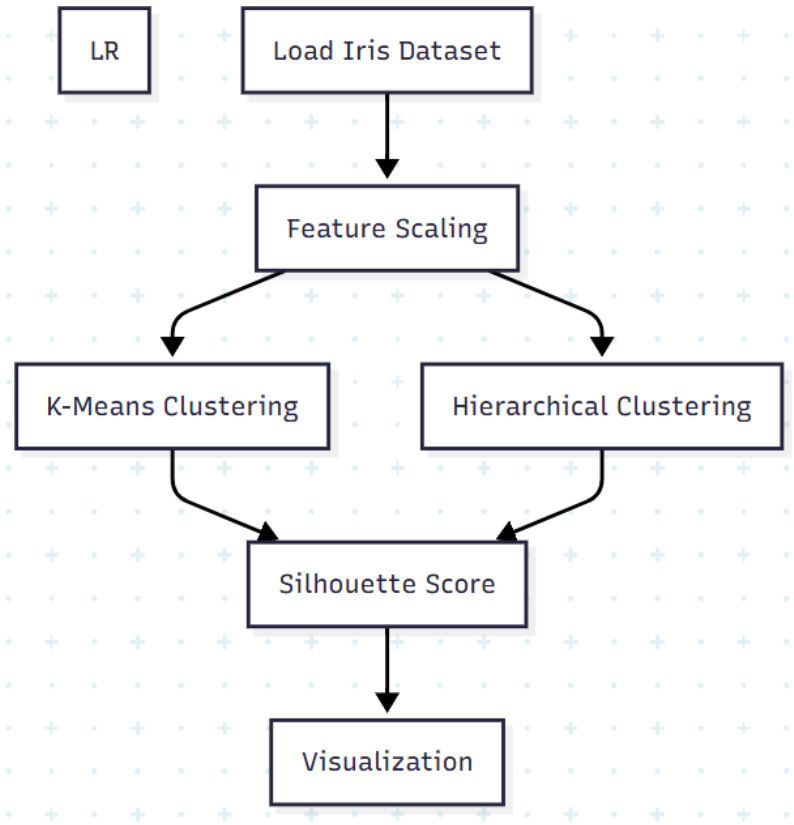
5. Methodology

1. Loaded Iris dataset
2. Applied feature scaling
3. Used Elbow method to determine optimal k

4. Applied K-Means clustering
5. Generated dendrogram
6. Applied Agglomerative clustering
7. Evaluated using Silhouette Score

WORKFLOW



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6. Performance Analysis

Silhouette Score:

$$S = \frac{b - a}{\max(a, b)}$$

Where:

- a = intra-cluster distance
- b = nearest cluster distance

Higher score indicates better clustering.

Both algorithms formed 3 clusters corresponding roughly to the 3 flower species.

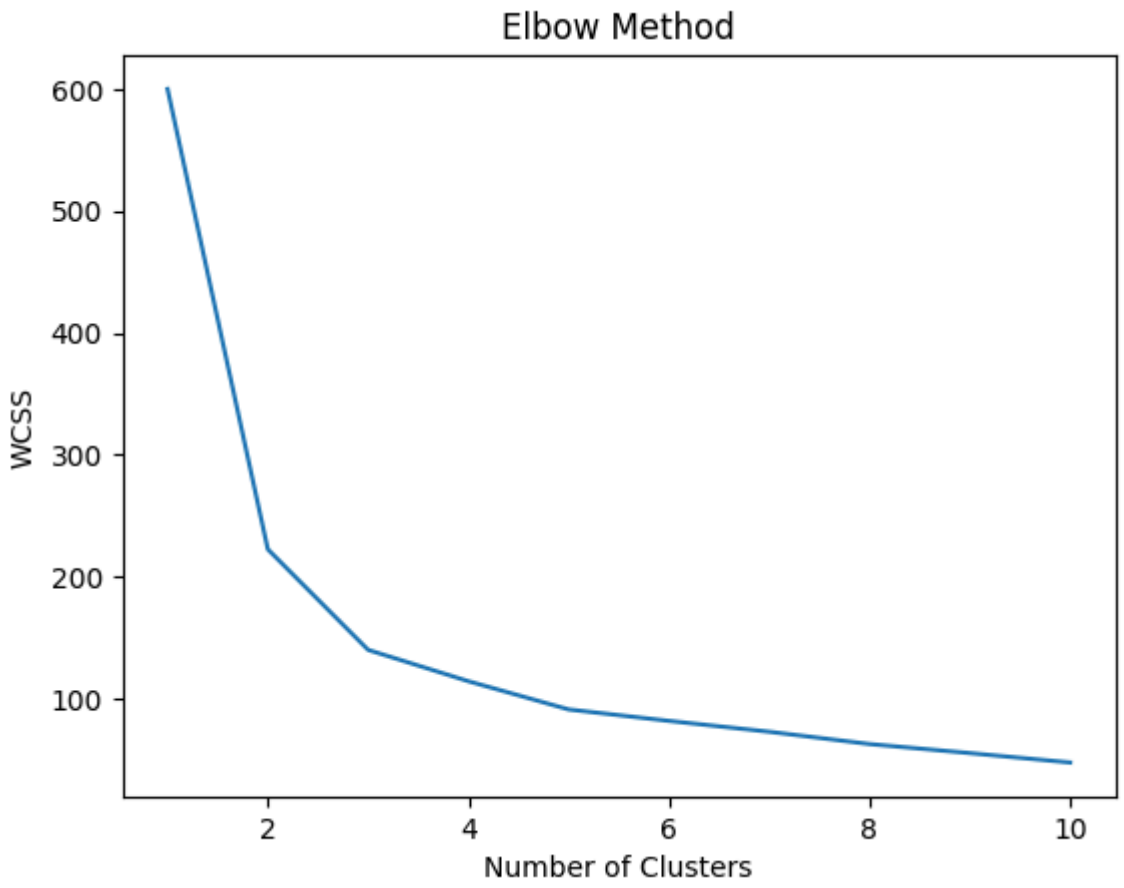
7.CODE

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.cluster import KMeans, AgglomerativeClustering
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
from scipy.cluster.hierarchy import dendrogram, linkage

# ----------------------------
# 1. Load Dataset
# ----------------------------
iris = load_iris()
X = iris.data
feature_names = iris.feature_names

print("Dataset Shape:", X.shape)

# ----------------------------
# 2. Feature Scaling
# ----------------------------
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# ----------------------------
# 3. K-Means Clustering
# ----------------------------

# Elbow Method
wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)
```

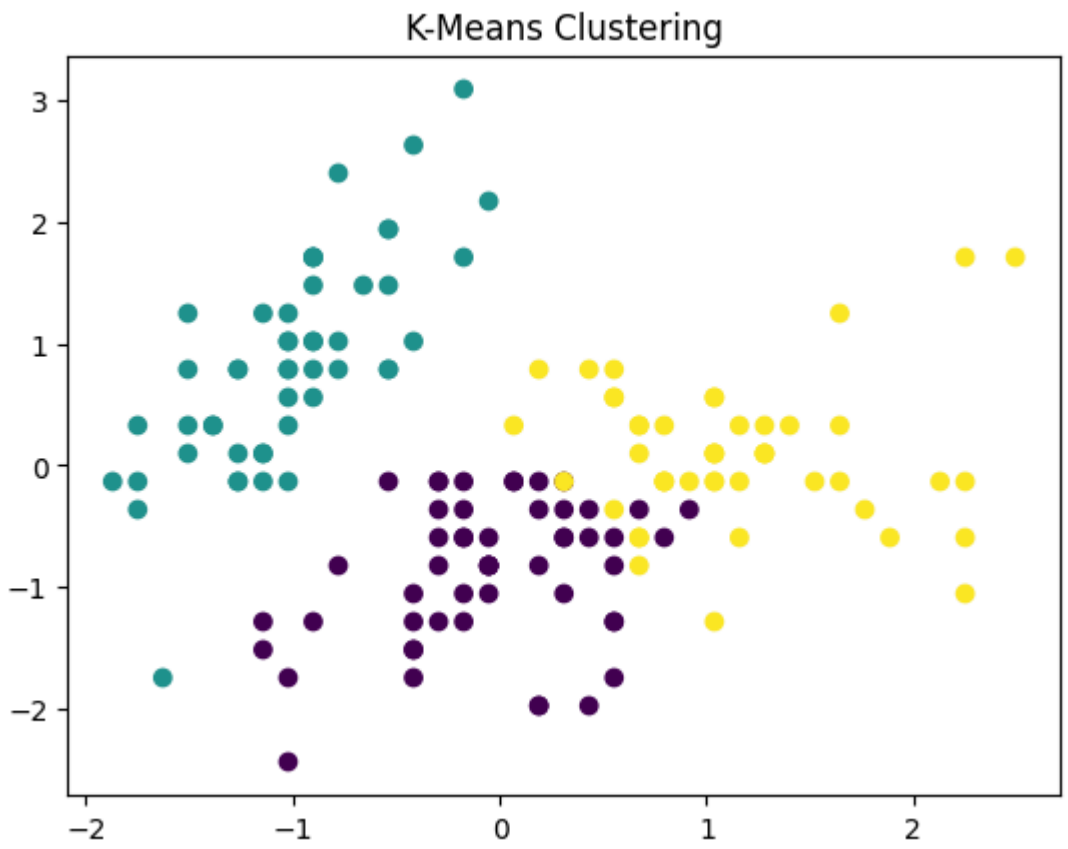
```
plt.figure()
plt.plot(range(1,11), wcss)
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()

# Apply K-Means (k=3)
kmeans = KMeans(n_clusters=3, random_state=42, n_init=10)
kmeans_labels = kmeans.fit_predict(X_scaled)

print("K-Means Silhouette Score:",
      silhouette_score(X_scaled, kmeans_labels))

plt.figure()
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=kmeans_labels)
plt.title("K-Means Clustering")
plt.show()

# ----------------------------
# 4. Hierarchical Clustering
# ----------------------------

plt.figure()
linked = linkage(X_scaled, method='ward')
dendrogram(linked)
plt.title("Dendrogram")
plt.show()

hc = AgglomerativeClustering(n_clusters=3)
hc_labels = hc.fit_predict(X_scaled)

print("Hierarchical Silhouette Score:",
      silhouette_score(X_scaled, hc_labels))

plt.figure()
plt.scatter(X_scaled[:,0], X_scaled[:,1], c=hc_labels)
plt.title("Hierarchical Clustering")
plt.show()

print("Experiment Completed Successfully ")
```

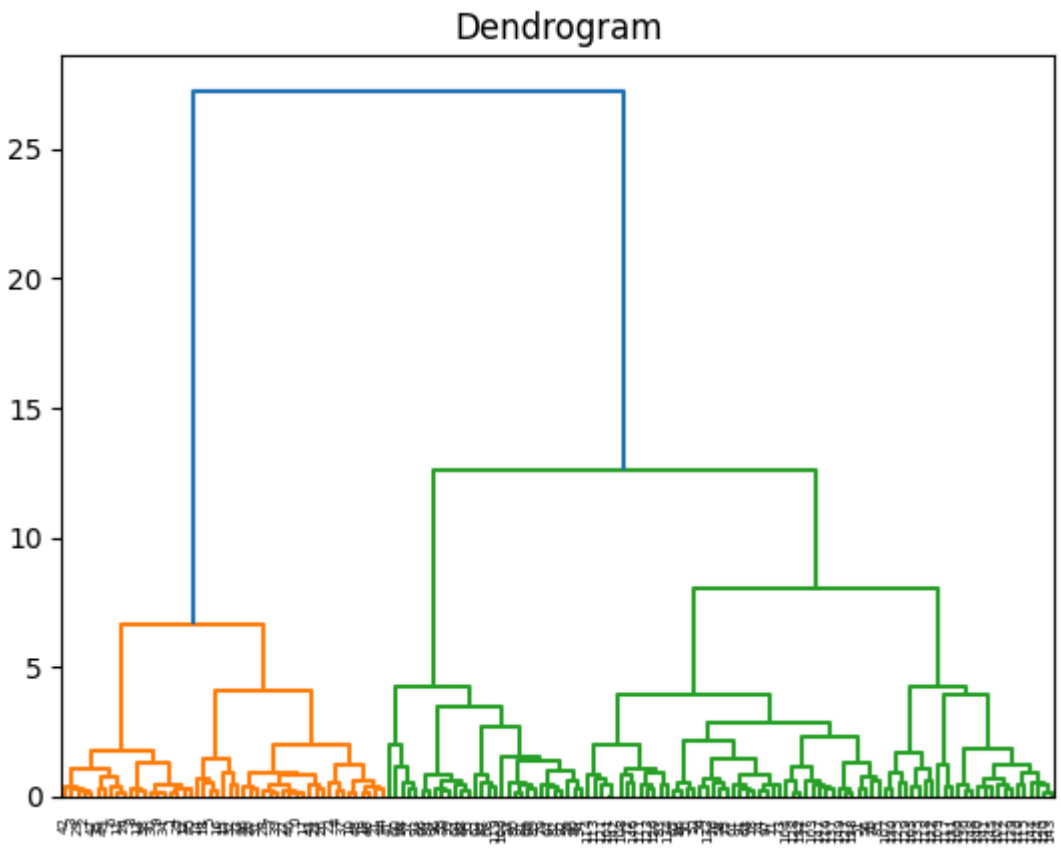
8.OUTPUT

Dataset Shape: (150, 4)

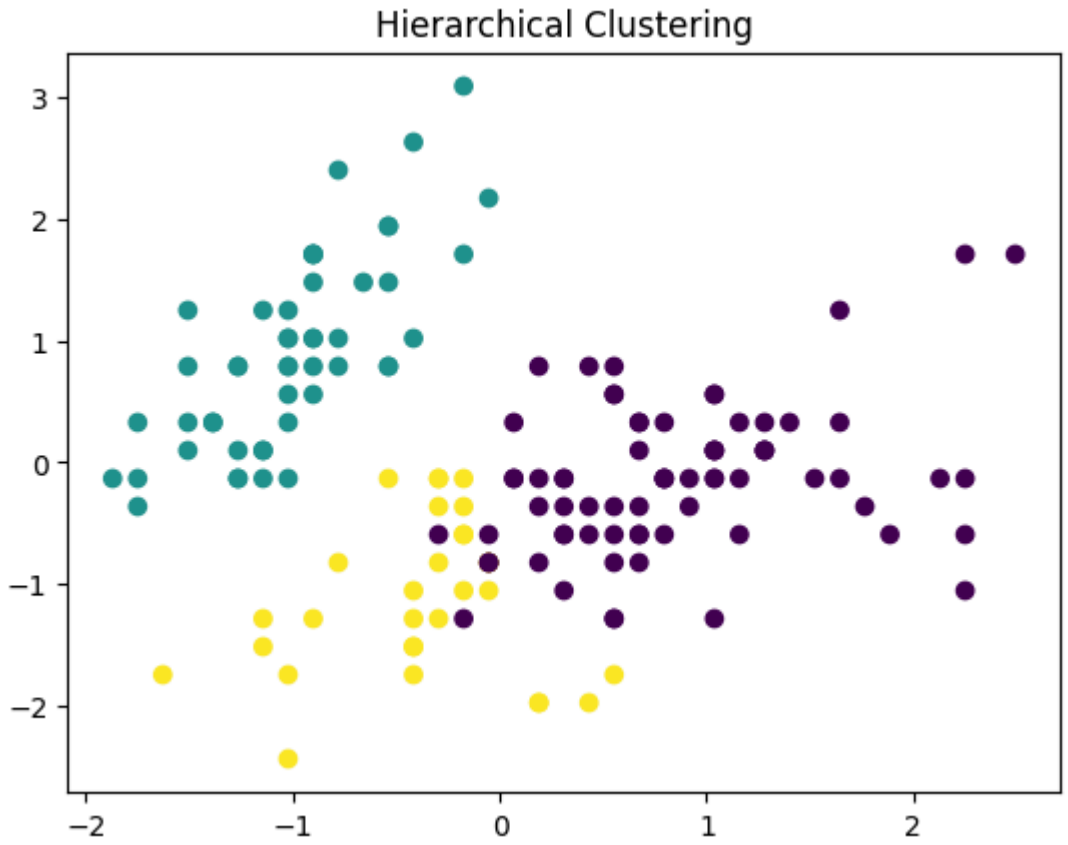


K-Means Silhouette Score: 0.45994823920518635





Hierarchical Silhouette Score: 0.4466890410285909



9.Conclusion

K-Means and Hierarchical clustering were successfully applied to the Iris dataset.

Both methods formed meaningful clusters.

Silhouette score indicated good cluster separation.

Thus, clustering techniques effectively grouped similar data points.